Capstone Project Documentation

Project Title
End-to-End Automation Testing of The Souled Store Website Using Playwright with JavaScript

Objective
The objective of this project is to design and develop an automated testing framework for The Souled Store e-commerce website using Playwright with JavaScript. This project focuses on automating major user workflows to improve testing efficiency, reduce manual effort, and ensure better application quality.

Modules Covered

1. Homepage Testing
Testing homepage elements such as navigation menu, banners, footer links, and homepage loading validation.

2. Search Functionality
Testing product search behavior using valid and invalid inputs along with search result validation.

3. Product Listing Page (PLP)
Testing product listing display, filters, sorting functionality, and product card validation.

4. Product Details Page (PDP)
Testing product information, image gallery, size selection, and add-to-cart functionality.

5. Cart Module
Testing add/remove product functionality, quantity updates, subtotal calculation, and cart validations.

6. Login Module
Testing valid and invalid login scenarios along with authentication validations.

7. Wishlist Module
Testing add/remove wishlist functionality and wishlist item management.

8. Checkout Flow
Testing checkout navigation, order summary validation, and checkout process flow.

Approximately 120 test cases will be designed and automated across these modules.

Technologies Used

Technology	Purpose
JavaScript	Programming Language
Playwright	Automation Testing Framework
Node.js	Runtime Environment
Visual Studio Code	Development IDE
Git & GitHub	Version Control
HTML/Allure Reports	Test Reporting

GitHub Actions	CI/CD Integration

Conclusion

This project provides hands-on experience in real-world automation testing using Playwright and JavaScript. By automating critical functionalities of The Souled Store website, the framework helps improve test coverage, execution speed, and application quality while following industry-standard automation practices.